

Session Objectives

1. Be able to classify which immune function tests are appropriate for each relevant immune cell type
2. Be able to describe the mechanism of how each immune function test works
3. Be able to compare and contrast screening from confirmatory testing
4. Be able to interpret immune function laboratory data findings
5. Be able to differentiate which immune function tests are appropriate for specific disease pathologies



When to be concerned?

History:

- Indicative of a familial origin
- Reason to believe in abnormal development
- Environmental changes
- Abnormal lab work in past: Anemias, Neutropenia
- Recurrent infections, Risk of HIV
- Patient's sex
- Failure to Thrive
- Fever of unknown origin

Physical Examination:

- Indicative of recurrent infections: *Candida albicans*
- Indicative of splenomegaly
- Multiple systemic features
- Check for lymph nodes

Prevalence of Immunodeficiencies

- Antibody Deficiencies: 55%
- Cellular/Humoral Deficiency: 15%
- Congenital defects: 10%
- Immune Dysregulation: 5%
- Autoinflammatory: 5%
- Complement: 4%
- Combined Immunodeficiencies: 3%
- Deficiency in innate immunity: 3%



Acute Phase Reactants

- Elevated in both acute & chronic inflammatory states
- Non-specific measure
- Indicative of general status of IS at the time
- Anything that increases by 25% during an inflammatory state
- **C-Reactive Protein (CRP):**
 - PRP
- **Erythrocyte Sedimentation Rate (ESR):**
 - Measures viscosity of blood
 - Increase in viscosity from fibrinogen break down



Acute Phase Reactants

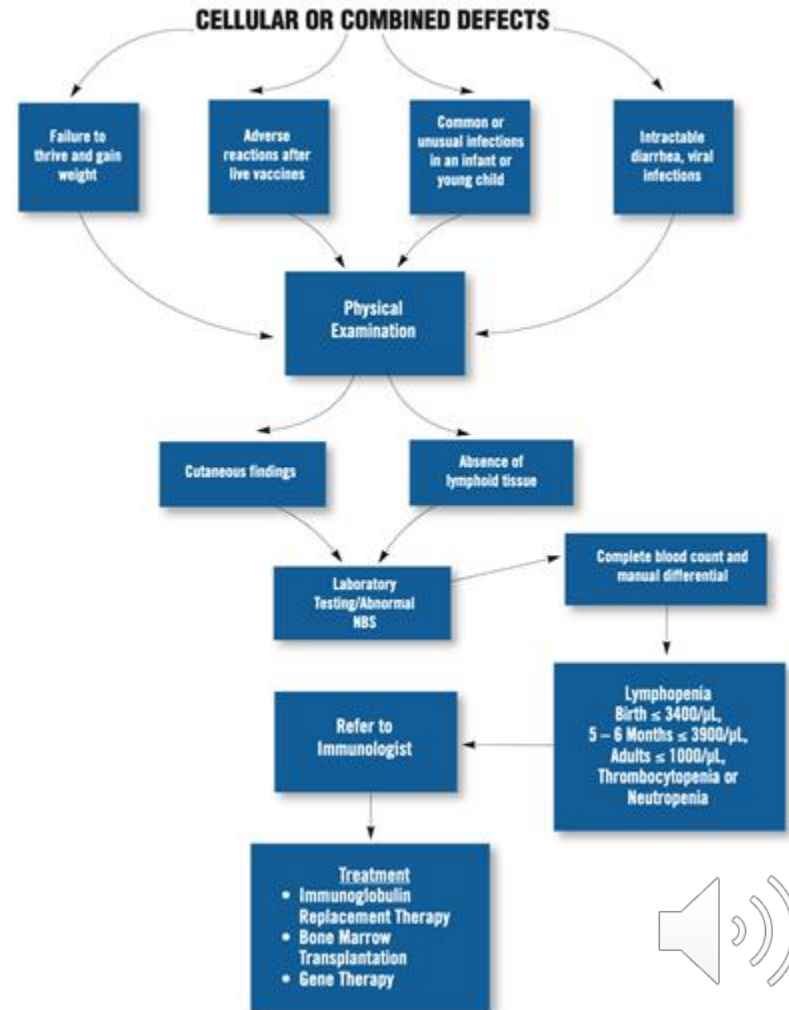
- **C-Reactive Protein (CRP):**
 - PRP
- **Erythrocyte Sedimentation Rate (ESR):**
 - Measures viscosity of blood
 - Increase in viscosity from fibrinogen break down
- Named after reaction to 'C' arbohydrate proteins from 'C'apsule of streptococcus
- Elevations in inflammation, infection, injury, occasionally in chronic illness, age determinant



Cellular Defects

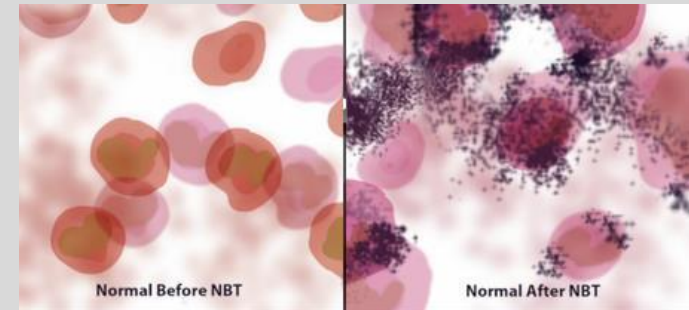
- Neutrophils: Innate
 - Phagocytosis with bactericidal enzymes
 - Provide extra support to macrophages
 - Derived from bone marrow
- NK cells: Innate
 - targets intracellular pathogens & tumors
 - IFN-Gamma to activate macrophages
- T-cells: Adaptive
 - CD4: regulator of adaptive IR
 - CD8: targets intracellular pathogens & tumors
- B-cells: Adaptive
 - Ab production

(Immune Deficiency Foundation, 2015)



Neutrophil Enumeration & Quality

- CBC w/diff & Absolute Neutrophil count:
Neutropenia:
 - Congenital: Severe congenital neutropenia
 - Iatrogenic: Cancer treatments
 - Infectious: Viral infections
- Dihydrorhodamine: DHR Flow Cytometry assay
 - Green fluorescence dye absorbed by cells
 - Chronic Granulomatous Disease
- Nitroblue Tetrazolium: NBT: NADPH Oxidase activity
 - Yellow to Blue by NADPH Oxidase
 - Chronic Granulomatous disease



(Liu, Lucy, 2021)





Causes of lymphocytopenia

Infection
Viral (eg. HIV, SARS, SARS-CoV-2, influenza, measles, hepatitis)
Bacterial (eg. tuberculosis, typhoid fever, brucellosis)
Fungal (eg. histoplasmosis)
Parasitic (eg. malaria)
Congenital immunodeficiency disorders
Iatrogenic
Immunosuppressive agents (eg. glucocorticoids, antilymphocyte globulin, alemtuzumab, rituximab)
Chemotherapy (eg. fludarabine, cladribine), hematopoietic cell transplantation
Radiation therapy (eg. total body irradiation, radiation injury)
Systemic disease
Autoimmune disorders (eg. systemic lupus erythematosus, rheumatoid arthritis, Sjögren syndrome)
Lymphoma
Other malignancies
Sarcoidosis
Renal failure
Aplastic anemia, pancytopenia
Cushing's syndrome
Other causes
Alcohol abuse
Zinc deficiency
Malnutrition, stress, exercise, trauma
Thoracic duct leak, rupture, diversion
Protein-losing enteropathy
Idiopathic CD4+ lymphocytopenia

HIV: Human immunodeficiency virus; SARS: Severe acute respiratory syndrome; SARS-CoV-2: SARS coronavirus 2.

Screening Tests:

- HIV: CD4+ ↓
- Thymic Aplasia: CD4+, CD8+ ↓
DiGeorge & Velocardiofacial syndrome
- Severe Combined Immunodeficiency: CD4+, CD8+ ↓
- Ataxia Telangiectasia: CD4+, CD8+ ↓

Lymphocyte Enumeration: **NK cells**

- Check Levels for CD16+/CD56+:
 - NK deficiency syndrome: No NK cells
 - SCID: T, B, & NK cells
 - Paroxysmal nocturnal hemoglobinuria: Pancytopenia
 - Ataxia Telangiectasia: ↑
- Consider testing in severe, recurrent, atypical HSV infection
- Deficiency leads to susceptibility to: VZV, HSV, EBV, & CMV



Lymphocyte Enumeration: T-cells

- Incredibly important for T-cell function & level to be intact
- Can lead to several combined pictures of immunodeficiency

Check Levels for:

- CD2+
- CD3+
- CD4+ & CD8+
 - Thymic Aplasia: secondary to DiGeorge's Syndrome
 - SCID: T, B, & NK cells
 - Ataxia-Telangiectasia: ATM gene



Lymphocyte Function: T-cells

- Stimulation of T-cells:
 - PHA: Phytohemagglutinin
 - Low stimulatory index, decreased T-cell function
 - Chronic mucocutaneous disease
 - Absent proliferation to candida antigen in vivo
- Cytokine production: IL-2, IFN-Gamma
 - IL-12R deficiency: decreased IFN-Gamma release
- TCR excision assay(TREC): circular DNA found in T-cells
 - SCID



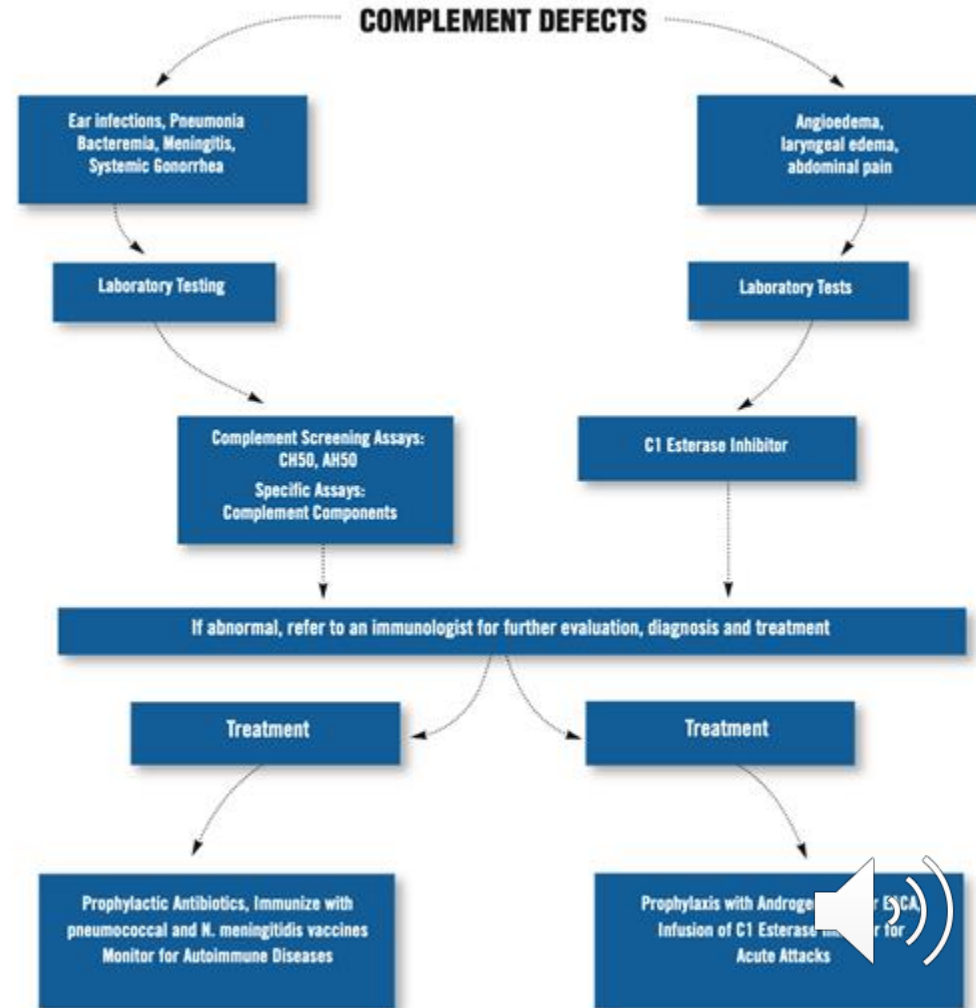
Lymphocyte Enumeration: **B-cells**

- Check Levels for CD19+, CD20+
- X-linked Brutons Agammaglobulinemia: Male predominant
 - No B-cell maturation: BTK gene
- SCID: T, B, & NK cells
- Ataxia Telangiectasia: ATM gene
- Pre-B Acute Lymphoblastic leukemia: Increased B-cells



Messenger Protein Abnormalities

1. Complement
2. Interleukins
3. Interferon



Complement Quantity defect

- **CH50:** Classical pathway, Hemolytic complement assay
 - non-specific, screening
- **AH50:** Alternative pathway: Hemolytic complement assay
 - Neisserial infection, most common indication
 - Determines deficiency in C3, C5-9
- **C1-C9 Enumeration**
 - SLE:** C1-C4 ↓
 - Neisseria* Bacteremia:** C5-C9 ↓



Complement Disorders

C1-Esterase inhibitor deficiency:

C4 ↓

-Hereditary angioedema

Further importance on history
taking and physical examination



Actemra/Tocilizumab

- MAB, monoclonal antibody
- IL-6 receptor inhibitor
- Used in treatment of inflammatory and autoimmune conditions
- Different types of arthritis
- COVID-19 use
- Manifests in alteration of the cytokine release pathways



Interferon Gamma Release Assay

- **Used to determine TB infection**
- **IGRA/QuantiFERON:** WBC release IFN-Gamma
 - Antigen-WBC mix
 - Levels tested for amount of IFN-Gamma released
 - Response: Determinate of LTB infection
 - Lab should reveal both qualitative & quantitative response



Ab Defects

- **Deficiency:**

- Sinus and Pulmonary infections
- Give way to infection by:
 - Streptococcus Pneumoniae*
 - Haemophilus Influenzae*
 - Other viral agents

- **Overproduction:**

- Multiple myeloma
- MGUS
- Waldenstrom Macroglobulinemia



Quantitative Evaluation

- **IgG** in response to an Ag
 - IS detects Ag
 - B-cells function
 - Invalid if patient received Ig in past 6 months
- **IgA:** IgA deficiency an extremely common cause of immune deficiency.
 - Seen in recurrent mucosal infections, breakdown in initial barrier to infection (MALT)
- **IgM:** Brutons agammaglobunemia
- **IgE:** Hyper-IgE (Job) syndrome & Type I HSN



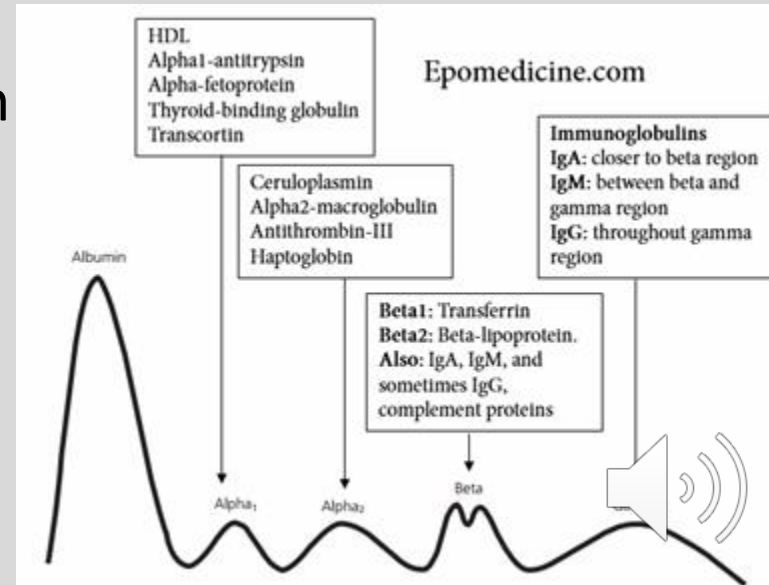
Quantitative Evaluation

- Wiskott-Aldrich syndrome: $\uparrow$ IgA, $\uparrow$ IgE; $\downarrow$ IgG, $\downarrow$ IgM
- AIDS: Increase in all Ig classes
- Ataxia-Telangiectasia: $\downarrow$ IgA, IgE, IgG
- SLE: Increase in all Ig classes
- RA: Increase in IgA or all
- Scleroderma: increase in all Ig
- Celiac Disease: Low IgA
- Primary biliary cholangitis: $\uparrow$ IgM



Quantitative Evaluation

- **Serum Protein Electrophoresis: SPEP**
 - M Protein of all Igs
 - Abnormal protein made that leads to pathology
- **Important in diagnosing:**
 - Multiple myeloma: IgG/IgA M-protein
 - MGUS:
 - <10% IgM M protein in Bone Marrow
 - Waldenstrom Macroglobulinemia:
 - >10% IgM M protein



Type 1 Hypersensitivities

- **IgE** reaction to external Ag
 - Specific to Ag presented
 - Can lead to anaphylaxis
 - Increased level seen in Atopic patients
 - Involved in type I HSN
- **Skin Testing**
- **Exogenous/Environmental:** Food, pollen, molds, animal dander, & insects



Autoimmune

Anti-Nuclear Antibody: ANA

Anti-dsDNA/Anti-Sm: SLE

Anti-MPO/P-ANCA: UC

Anti-U1 RNP: MCTD

Anti-SSA/SSB: Sjögren

Anti-Scl70/Anti-DNA Topoisomerase: Scleroderma

Anti-Cardiolipin: SLE, Anti-phospholipid syndrome

Anti-CCP:RA

Anti-CD55/DAF: Paroxysmal nocturnal hemoglobinuria

Anti-Thyroid peroxidase: Hashimoto Thyroiditis

Anti-Parietal Cells: Pernicious anemia

Anti-Endomysial: Celiac disease

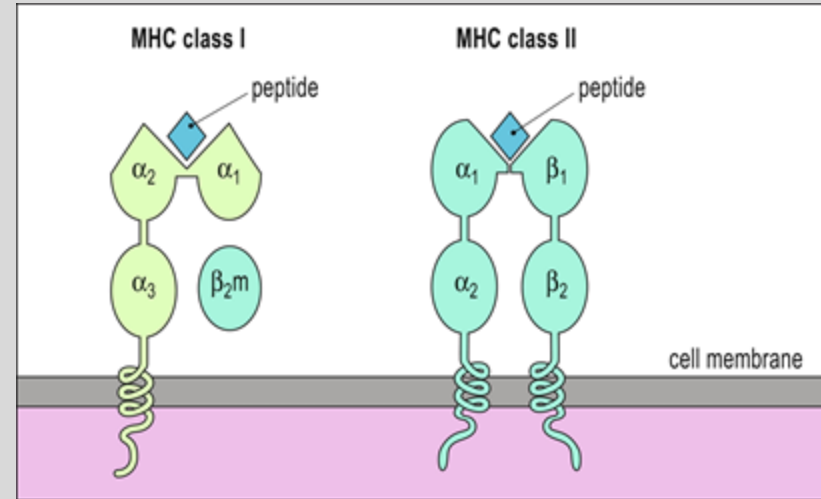
Anti-Smooth Muscle: Autoimmune hepatitis type 1

Anti-Mitochondrial: Primary Biliary cholangitis



Human Leukocyte Antigen

- **MHC-I HLA:** Self Ag
 - HLA-A, B, & C
 - Important in transplantation
 - HLA-DR2: MS, SLE, Goodpasture
 - HLA-DR3: DM-I, SLE, Graves
 - HLA-DR4: RA, DM-1, Addison's
- **MHC-II HLA:** Foreign Ag
 - HLA-DR, DQ, & DP
 - Associated with autoimmune:
 - HLA-B27: Psoriatic arthritis, Ankylosing Spondylitis, IBD-associated arthritis, reactive arthritis



(Male, et al., 2021)

